UC-B Campus Marketplace with Reviews

Week 8 - Project Proposal (approved; corrected 7 November 2026)

CPE 314 - Software Design

Group Name / No.: Group 1 / Team 01 - BSCPE 3B - EDP 69323

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Theme: Campus Marketplace with Reviews

API: FakeStoreAPI - https://fakestoreapi.com

Version: V 0.2 - instructor corrections, 7 November 2026

1. One-Paragraph Pitch

UC-B Campus Marketplace with Reviews is a local Java desktop app for students, faculty and school groups to post and browse campus goods. Textbook, Electronics and Furniture listings have their own condition rules. Recursive category search, buyer-interest queues, recently viewed stacks, reviews, SQLite/JDBC persistence, public API suggestions, background JSON sales-history import and category charts demonstrate the required lessons. The prototype uses demo members; real login and online payments are outside its scope. The proposal was approved after the Week 8 revision.

2. Entity-Relationship Sketch

The revised ERD shows all ten application tables and all their attributes. Listings is repeated across three views to keep the diagram readable. PK = primary key, FK = foreign key, UK = unique key. Complete SQL and constraints are in Section 4 of the project design document.

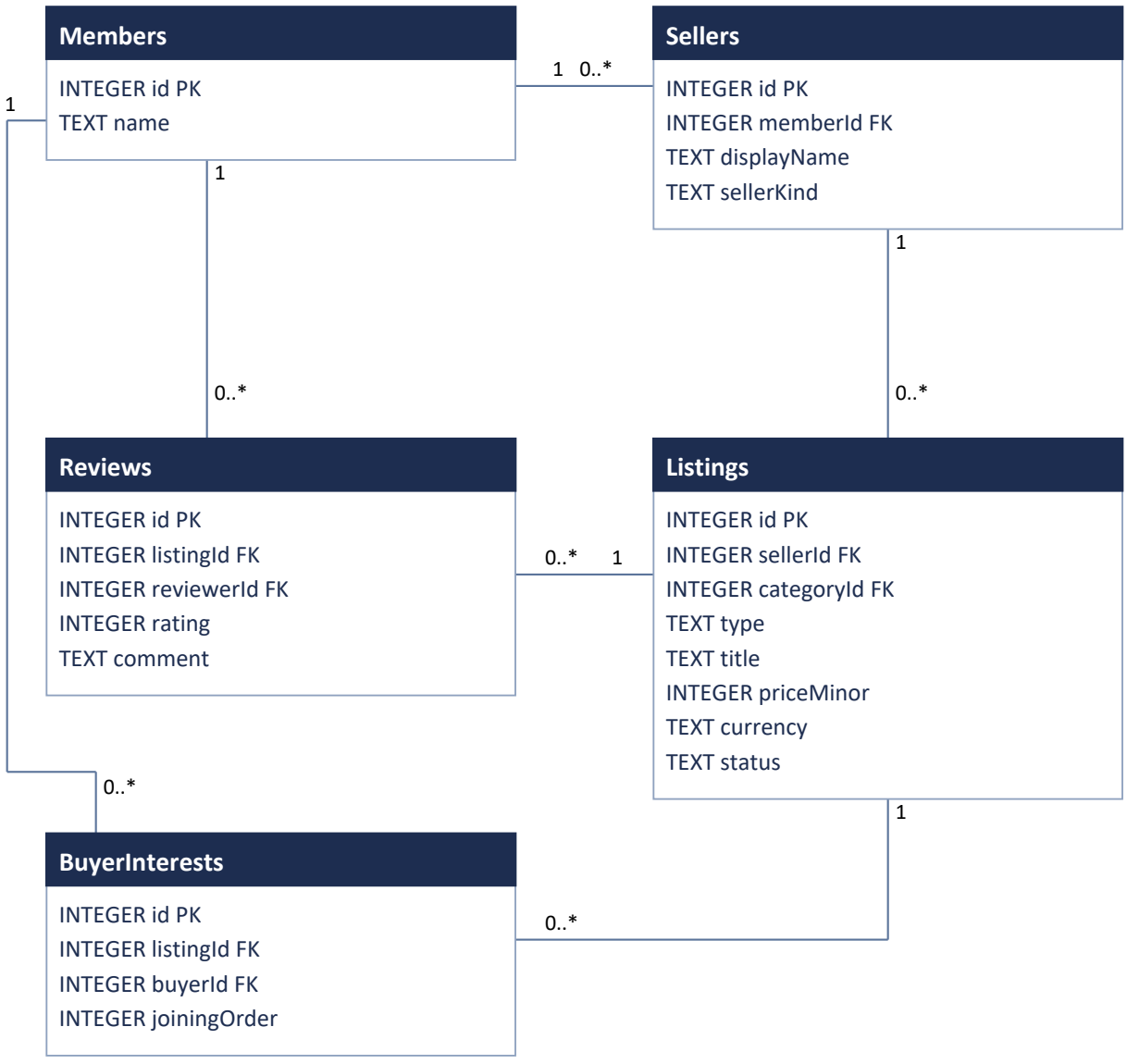


Figure 1: Members, sellers, listings, reviews and buyer interests

Categories and imported sales

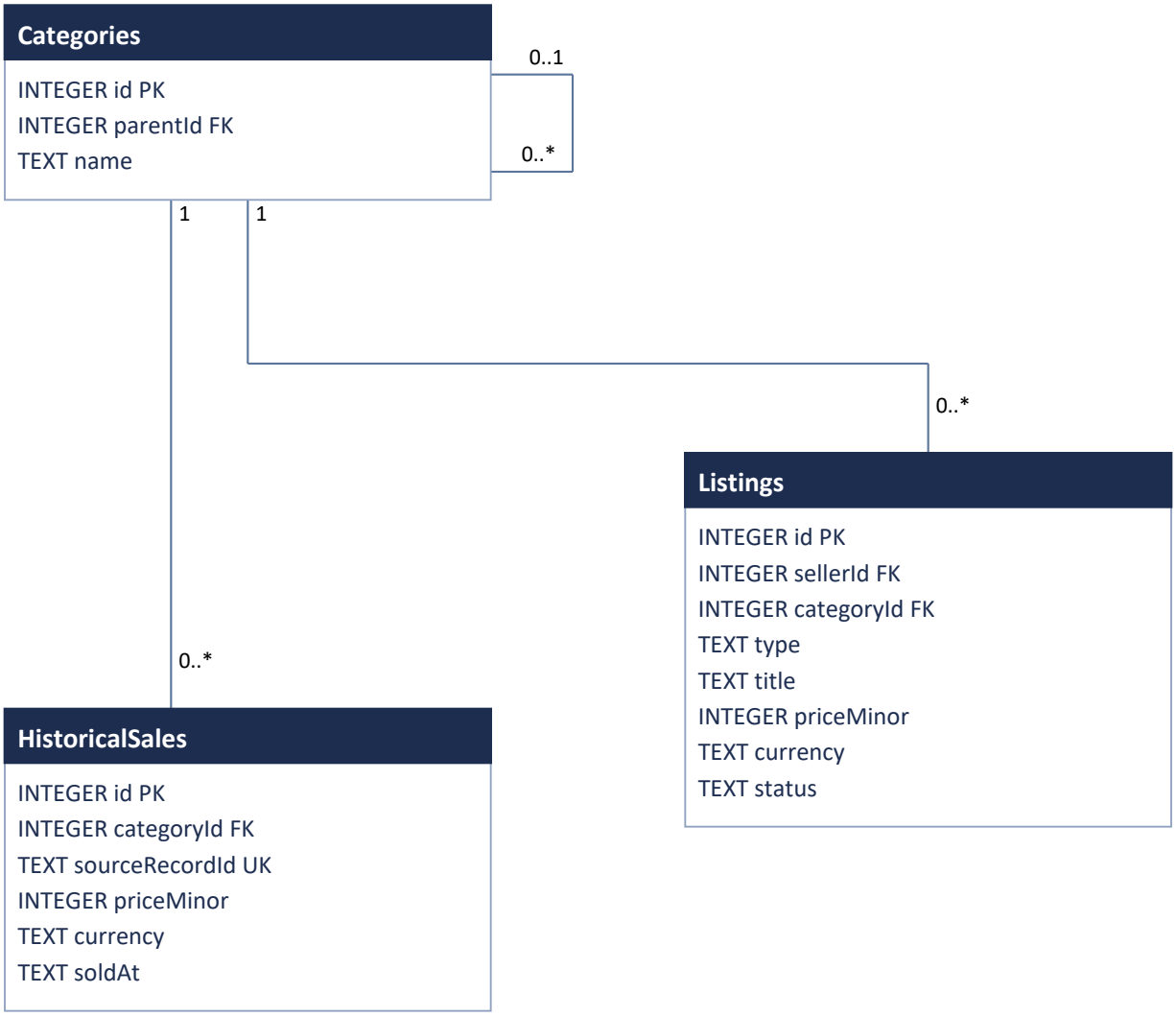


Figure 2: Categories, listings and historical sales

Each category has zero or one parent and zero or more children. Current listings and historical sales each belong to one category.

Condition details



Figure 3: Type-specific listing details

Each listing has exactly one matching subtype-details row overall. listingId is both the PK and FK in each detail table. Transactions and Java validation enforce the matching-type rule.

3. Feature Checklist

Table 1: Features and course requirements

Feature	Requirement
Abstract Listing with three condition-grading subclasses	Object-oriented design and polymorphism
FIFO waitlist, LIFO recent history and recursive category search	Data structures and recursion
JUnit normal and edge-case checks	Automated testing
Related SQLite tables and PreparedStatement persistence	Database modeling and JDBC
Live FakeStoreAPI catalog suggestions	API consumption
JavaFX controls and background JSON/API operations	Event-driven GUI and concurrency
Category means, counts and labeled bar charts	Data mining and visualization

4. Task Split

Table 2: Team responsibilities

Member and role	Responsibility
John Benedict Alaiza - Backend Developer (Listing Models)	Listing classes, condition rules and validation.
Acexion B. Mahigugmaon - Project Leader / Integration	Connect the components, maintain documents and coordinate fixes.
Glyde V. Arreglado - Database Admin	SQLite tables, saved records and summary queries.
Tristan Dominic Urot - UI/GUI Developer	JavaFX controls, event handlers and charts.
Joshua Brian S. Enovero - Backend & API Developer	Catalog requests, JSON parsing and import.
Emanuelle De Paz - QA & Testing	Test cases, expected calculations and error checks.

5. Public API

FakeStoreAPI GET https://fakestoreapi.com/products returns product JSON without an API key. Electronics title matching supplies category and catalog reference-price suggestions. API currency is unspecified; the member enters the final PHP price. Unsupported or unmatched items keep manual entry available. See API documentation.

6. Risks / Open Questions

1. If FakeStoreAPI matches a new electronics product but gives no currency, how will we prevent a misleading PHP resale-price suggestion?

2. If a listing has several campus reviews, how will we stop its price from being counted several times in the category price average?

3. If the same sales JSON file is imported twice, or cancellation occurs after some rows are processed, how will HistoricalSales avoid duplicate or partially saved sales?

4. If a Listings row loses its TextbookDetails, ElectronicsDetails or FurnitureDetails record, how will its owner restore the actual condition without replacing the listing or its reviews?